Modulation of Inhibitory Control Across High and Low Energy Food in Healthy Adults

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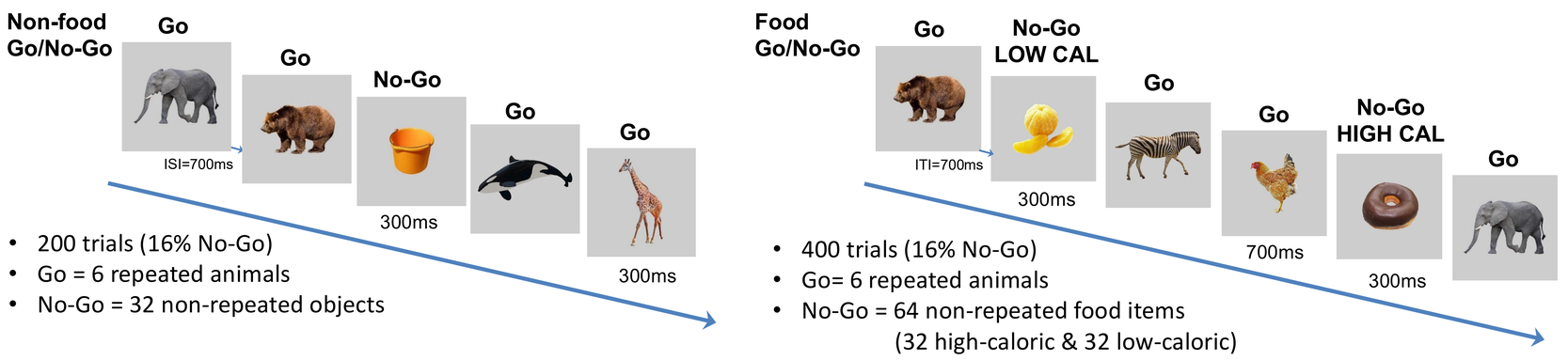
INTRODUCTION

- Maintaining healthy weight requires balancing food consumption and energy expenditure. Unfortunately, increased availability of food, especially high energy (calorie dense) foods, increases appetitive drive and challenges inhibitory control in dietary decision making. Understanding the neural basis of inhibitory control for food cues could inform mechanisms underlying obesity and other food-related disorders.
- Inhibitory control has been assessed by a variety of tasks which reveal an individual's ability to suppress prepotent responses. Inhibition is reflected in behavioral outcomes as response accuracy.
- In a food Go/No-Go task there were more commission errors with shorter RT to food cues compared to nonfood cues [1].
- The N2 and P3 of the ERP in Go/No-Go tasks are assumed to represent inhibitory control processes [2]. ERP studies of food Go/No-go tasks varied in their outcomes due to methodological differences. [3],[4].
- We created a modified food Go/No-Go in which the Go stimuli were 6 repeated animal pictures and the No-Go images were either non-repeated non-food, or non-repeated high and low caloric food images. We hypothesized that No-Go food cues will engage inhibitory control process more than non-food cues that will be reflected in N2 amplitude. We also expect larger N2 amplitude for the high calorie foods compare with low calorie foods.

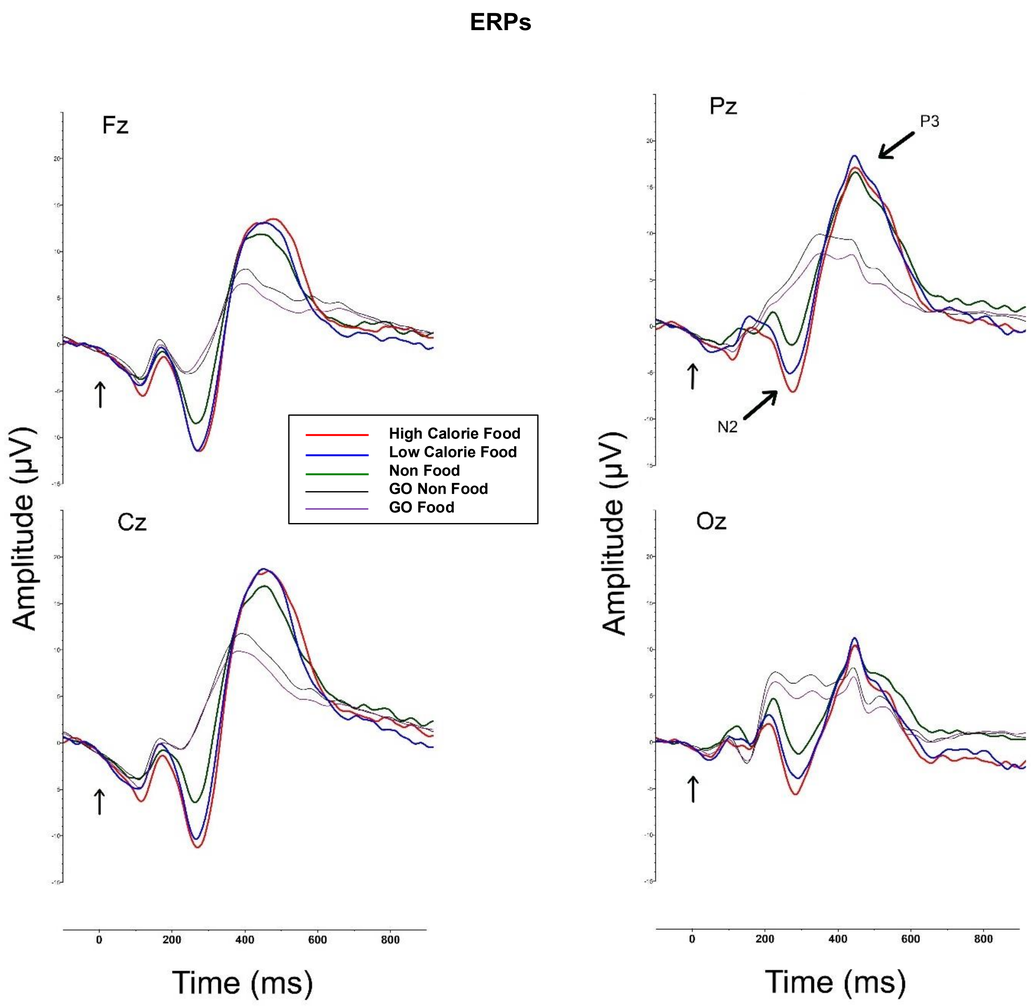
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GO / NO-GO TASK



RESULTS



	False Alarms %	Hit RT (ms)	False Alarms RT (ms)
Non-food	15.63	288	256
Food	33.44	277	248
High Calorie	35.00		
Low Calorie	31.88		

- %FA for non- food vs. food is significant (Pair t-test, P<0.001), but there is no difference between high and low calorie food.
- RT for Hit was significantly slower than RT to FA (P<0.001), but RT of FA in the two Go/No-Go tasks did not differ. In addition, RT of FA for the high calorie did not differ from RT of FA for the low calorie food.
- Statistical comparison of the N2 amplitudes at Fz, Cz, Pz and Oz show that food had significantly larger negative amplitude than non-food. Low calorie food elicited N2 amplitude that was qualitatively smaller than the one elicited by the high calorie food, but larger than that of non-food. However it was not statically different from either of them.
- Statistical comparison of the P3 amplitudes at Fz, Cz, Pz and Oz, showed that the P3 amplitude did not differ between the 3 conditions.

METHODS

Participants:
- 12 healthy young-adults participated in the study, but data from 2 participants was discarded because of chance performance (FA >40 %). Mean age of the 10 participants was 25.4 years, 7 were males, and the mean BMI was 21.58 kg/m^2. Subjects recruited into the study if they reported no dietary restrictions or excessive exercise regimen.

Material:
- Pictures were selected from both the Food-Pictures database [5] and from the internet.
- Stimuli were presented on a screen at a distance of approximately 1 m from the subject.

EEG recordings and preprocessing:
- EEG at 64 scalp sites (10-20 system) was recorded using ActiCAP (Brain Product) with nose as reference, with 500 Hz sampling rate. Bipolar VEOG and HEOG was recorded to monitor eye movements.
- Data processing included filtering, artifact removal by visual inspection, and ICA was used for ocular correction (Vision Analyzer).

ERP analysis:
- EEG data were segmented to 1000 s epochs with 100 ms baseline correction pre-stimulus. ERP were generated by averaging epochs for each stimulus type. N2 and P3 amplitudes were measured by computing voltages averages between 240 -300 ms, and between 400 – 500 ms, respectively in 4 midline electrode sites (Fz, Cz, Pz, Oz). Repeated Measure ANOVA was used to compare N2 and P3 amplitudes between stimulus conditions.

CONCLUSIONS

- The findings of larger FA and increased N2 amplitude for food images as compared to non-food images, indicate that food images demand more inhibitory control processes than non food cues. Qualitatively, larger N2 amplitude is seen for high compared to low calorie food suggesting that palatable food may require more cognitive control. However, this was not substantiated, possibly due to the small sample size.
- P3 amplitudes for food and non-food images were not significantly different, although P3 amplitudes at Fz and Cz show qualitative differences.
- **These data support the notion that inhibiting responses to food is more demanding than inhibiting responses to non-food. Thus, it is possible that impairment in inhibitory control process contributes to the difficulties in controlling food intake in an environment with readily available high calorie foods.**

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